Weil positivity in compact windows: certified two-sided bounds and a Landau–Widom decay law

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Abstract

Weil's criterion equates the Riemann Hypothesis with the positivity of an explicit quadratic form $Q(f)$. For test functions supported in a window $[-L, L]$ we study the profile $\lambda^*(L) = \inf Q(f)/\|f\|_2^2$ from both sides. A one-stroke reduction converts window positivity into positive semidefiniteness of a finite matrix; executing it at $L = 0.8$ yields the unconditional certificate $Q(f) \geq 8.9 \times 10^{-18}\|f\|_2^2$ for all (complex) f of support 1.6, extending the classical range $\log 2$ of Yoshida and Connes–Consani by a factor 2.3. The same computation shows that the window ground state is simple and even. On the other side, interval-arithmetic Rayleigh quotients, evaluated on the geometric side of the explicit formula with no appeal to zeros or to RH, give unconditional upper bounds down to 3.2×10^{-283} at $L = 2$. They follow $-\ln \lambda^*(L) \simeq 2\pi^2 N(T^*)/\ln N(T^*)$ with $T^* = 2\pi e^{2L}$, the Landau–Widom plunge rate for time–band limiting. At $L = 0.8$ the two halves enclose the profile, $8.9 \times 10^{-18} \leq \lambda^*(0.8) \leq 2.27 \times 10^{-17}$. Any pointwise-envelope certificate must resolve frequencies up to $T_1 = 2\pi e^{A_L}$ with $A_L \sim 4e^L$, a doubly exponential barrier. Under RH we prove $\lambda^*(L) \leq \exp(-Le^L)$ for all large L.

1 Introduction

Let $\Lambda(n)$ be the von Mangoldt function and let $\gamma_1 \leq \gamma_2 \leq \cdots$ be the ordinates of the non-trivial zeros of ζ in the upper half-plane. For an even real continuous test function f of compact support write $\hat{f}(r) = \int_{\mathbb{R}} f(u)e^{iru}\,du$. The Riemann–Weil explicit formula expresses the sum of $\hat{f}$ over zeros in terms of arithmetic and archimedean data, and Weil's criterion [1, 4] asserts that RH is equivalent to $Q(f) \geq 0$ for all admissible f. Every spectral approach to RH, from Hilbert–Pólya to the trace-formula program of Connes [5] and the prolate-operator program of Connes–Consani and Connes–Moscovici [6, 7], has to establish some variant of this positivity on larger and larger spaces of test functions.

Positivity is a yes/no statement. We ask instead how small $Q(f)$ can get, relative to $\|f\|_2^2$, on a fixed compact window, and introduce

$$\lambda^*(L) \;=\; \inf_{\substack{f \text{ even, real} \\ \operatorname{supp} f \subset [-L,L]}} \frac{Q(f)}{\|f\|_2^2}. \tag{1}$$

Under RH the zeros side gives $Q(f) = \sum_{\gamma>0} 2|\hat{f}(\gamma)|^2$, and a sampling argument (Proposition 4) yields $\lambda^*(L) > 0$ for every L. Conversely, a negative value at some L would disprove RH, and an

unconditional proof of $\lambda^*(L) > 0$ is a finite fragment of RH. The map $L \mapsto \lambda^*(L)$ is thus a difficulty profile of RH as seen through compactly supported test functions.

This note determines the profile from both sides: the reduction, the statements, and the numbers.

Lower bounds. Write the geometric side, for $\operatorname{supp} f \subseteq [-L, L]$ and $F = \hat{f}$, as

$$Q(f) \;=\; 2F(i/2)^2 + \frac{1}{2\pi}\int_{\mathbb{R}} |F(t)|^2\, \Psi_L(t)\, dt, \tag{2}$$

with the Weil symbol

$$\Psi_L(t) \;=\; \operatorname{Re}\,\psi\Big(\tfrac{1}{4} + \tfrac{it}{2}\Big) - \log\pi - \sum_{\log n < 2L} \frac{2\Lambda(n)}{\sqrt{n}}\cos(t\log n), \tag{3}$$

$\psi = \Gamma'/\Gamma$. Only prime powers with $\log n < 2L$ contribute. Let $A_L = \sum_{\log n<2L} 2\Lambda(n)/\sqrt{n}$ be the mass of the prime comb. The classical record is positivity for $2L \le \log 2$ [3, 6]. Theorem 7 removes the infinite-dimensionality of the window problem in one step: a pointwise envelope for Ψ_L, with a comb constant that Weyl equidistribution shows to be optimal, converts positivity on $[-L, L]$ into positive semidefiniteness of a single finite matrix. Executing it at $L = 0.8$ gives:

Theorem 1 (Certified positivity, support 1.6)**.** *For every real even $f \in L^2(\mathbb{R})$ with $\operatorname{supp} f \subseteq [-0.8, 0.8]$,*

$$Q(f) \;\ge\; 8.9 \times 10^{-18}\, \|f\|_2^2.$$

In particular the Weil form is positive on all autocorrelations $g = f \star \tilde{f}$ with $\operatorname{supp} g \subseteq [-1.6, 1.6]$.

The same reduction on the odd sector, where the pole term changes sign, yields $Q \ge 8.2 \times 10^{-15}$ for odd f, hence positivity for arbitrary complex test functions of support 1.6 (Corollary 9), and shows that the window ground state is simple and even (Theorem 8). This is the first of the two missing spectral hypotheses in the operator program of Connes–Consani–Moscovici [9] and Connes–van Suijlekom [8], verified here at a fixed scale with explicit constants.

Upper bounds. The same geometric side, evaluated at explicit trial functions in interval-closed multiprecision arithmetic, supplies unconditional variational upper bounds for $\lambda^*(L)$, from 10^{-6} at $L = 0.5$ down to 3.2×10^{-283} at $L = 2$ (Table 1). No zeros and no RH enter the certificates. The decay is governed by the resolution height $T^*(L) = 2\pi e^{2L}$, the height at which the density of zeta zeros saturates the Paley–Wiener budget L/π of the window. Writing $N(T)$ for the zero-counting function, the certified bounds follow

$$-\ln \lambda^*(L) \;=\; C\,\frac{N(T^*)}{\ln N(T^*)}\,(1 + o(1)), \qquad C = 20.13\ldots \approx 2\pi^2, \tag{4}$$

the Landau–Widom plunge rate for time–band limiting operators [13, 12]. A one-parameter fit to $L = 1.4, 1.6, 1.8, 2.0$ has residuals below 0.7%. Out of sample, with both models calibrated on $L \le 1.2$, the Landau–Widom residuals are flat ($\approx -5\%$, no trend), while those of a competing three-parameter "one nat per zero" model grow from $+0.8\%$ to $+5.8\%$. The constant is fitted, not derived; a rival normalization $4\pi\, e^{2L}$ is numerically entangled on this range and is not separated by these data. Under RH a theorem captures the qualitative content of (4) with exponent e^L in place of e^{2L}:

Theorem 2. *Assume RH. There is an L_0 such that for all $L \ge L_0$,*

$$\lambda^*(L) \;\le\; \exp\!\big(-L\, e^L\big).$$

Enclosure and barrier. Where the two programmes overlap they enclose the profile from both sides, with certificates on both sides:

$$8.9 \times 10^{-18} \;\le\; \lambda^*(0.8) \;\le\; 2.27 \times 10^{-17},$$

a factor 2.6. The same computations quantify why the positivity route, unassisted, cannot reach RH.

Theorem 3 (Barrier)**.** *Any application of Theorem 7 requires $T^\sharp > T_1(L) = 2\pi e^{A_L}$, and $A_L = (4 + o(1))e^L$ as $L \to \infty$ by the prime number theorem. The matrix size $N \asymp LT_1$ therefore grows doubly exponentially in L. Moreover $\sup_t$ of the prime comb equals A_L exactly, so within the class of certificates that bound the comb pointwise the threshold T_1 cannot be lowered.*

To improve on pointwise envelopes one must prove that the phases $(t \log p)_p$ cannot align—an arithmetic statement about $\{\log p\}$, of the same nature as the information carried by the zeros themselves.

Related work. The tradition of attacking RH-adjacent quantities numerically is represented by the de Bruijn–Newman constant [14]. Weil positivity for restricted classes of test functions was established by Connes–Consani using prolate spheroidal wave functions [6]; Theorem 1 extends the range of unconditional positivity itself. Finite-rank truncations of the Weil form have recently become objects of study [8, 9, 10, 11]. Those works study a fixed truncation family; the window-variational infimum (1), its unconditional certification, and its asymptotic law in L appear not to have been considered before.

2 The Weil form

For f as above, with $g = f \star \tilde{f}$, the geometric side of the explicit formula (normalization of [4]; see also [15, Ch. 5] and Barner [2]) is the identity (2). It is identified with the zeros side

$$Q(f) \;=\; \sum_{\rho} \widehat{g}\Big(\frac{\rho - \frac{1}{2}}{i}\Big) \;\overset{\text{RH}}{=}\; \sum_{\gamma > 0} 2\,\big|\hat{f}(\gamma)\big|^2, \tag{5}$$

and $Q(f) \ge 0$ for all admissible f if and only if RH holds [1]. Both halves of this note work with the geometric side. The zeros side is used only to propose trial vectors for the upper bounds, and as an independent cross-check; it never enters an unconditional certificate.

The zeros side is a sum of non-negative terms; the geometric side computes the same tiny number as a difference of large terms. When $Q(f) \approx 10^{-20}\|f\|^2$, double-precision evaluation of (2) is meaningless and readily produces spurious negative eigenvalues. Multiprecision interval arithmetic removes the obstruction for upper bounds. Lower bounds continue to require one-sided, sign-definite estimates on the geometric side.

For any finite-dimensional subspace V of admissible test functions, the least eigenvalue of Q on V is an upper bound for $\lambda^*(L)$. All numbers in Table 1 are of this form.

Positivity of the infimum under RH needs a short argument: pointwise positivity of Q does not by itself exclude $\inf_{\|f\|=1} Q(f) = 0$.

Proposition 4. *Assume RH. For every $L > 0$ there is $c_L > 0$ such that $Q(f) \ge c_L\|f\|_2^2$ for all admissible f supported in $[-L, L]$; in particular $\lambda^*(L) > 0$.*

Sketch. The transform $\hat{f}$ lies in the Paley–Wiener space PW_L^2. By Littlewood's theorem (a consequence of RH) the gaps $\gamma_{n+1}-\gamma_n$ tend to zero, so a maximal δ-separated subset Λ of $\{\pm\gamma\}$ with $\delta=\pi/(4L)$ has lower Beurling density $D^-(\Lambda)>L/\pi$. Beurling's sampling theorem [17] then yields $\sum_{\lambda\in\Lambda}|\hat{f}(\lambda)|^2 \ge c\,\|\hat{f}\|_2^2$, and the zeros-side identity gives the claim. □

No effective c_L comes out of the argument. An effective unconditional value is exactly Theorem 1.

3 One-stroke reduction

Lemma 5 (Crude envelope)**.** *For all $t \ge \frac{15}{4}$,*

$$\operatorname{Re}\,\psi\Big(\tfrac14+\tfrac{it}{2}\Big)-\log\pi \;\ge\; \log\frac{t}{2\pi}-\frac{1}{t}, \qquad \textit{hence} \qquad \Psi_L(t) \;\ge\; \log\frac{t}{2\pi}-\frac{1}{t}-A_L.$$

Proof. By $\psi(z)=\psi(z+1)-1/z$ it suffices to bound $\psi\big(\frac54+\frac{it}{2}\big)$. Binet's second formula gives, for $\operatorname{Re} z>0$, $\psi(z)=\log z-\frac{1}{2z}-2\int_0^\infty \frac{u\,du}{(u^2+z^2)(e^{2\pi u}-1)}$. At $z=\frac54+\frac{it}{2}$ one has $|u^2+z^2|\ge 5t/4$, and the integral is $\le 1/(15t)$. The remaining elementary estimates contribute at most $7/(2t^2)$, so the total deficit is $\le 1/t$ for $t\ge 15/4$. The comb is bounded by A_L trivially. □

Lemma 6 (Optimality of the crude constant)**.** *Write $P_L(t)=\sum_{\log n<2L}(2\Lambda(n)/\sqrt{n})\cos(t\log n)$. Then $\sup_{t\in\mathbb{R}}P_L(t)=A_L$. Consequently no pointwise bound $P_L(t)\le C$ valid for every t can improve on $C=A_L$.*

Proof. $P_L(t)\le A_L$ is trivial, with equality at $t=0$. That the supremum is approached at arbitrarily large t follows from Weyl equidistribution: $\{\log p\}$ are linearly independent over $\mathbb{Q}$ by unique factorization, so the phases $(t\log p)_p$ equidistribute on the torus and can be sent simultaneously arbitrarily close to 0. □

Theorem 7 (One-stroke reduction)**.** *Let $L>0$ and set $T_1:=2\pi e^{A_L}$. Fix any $T^\sharp$ for which*

$$\beta^* \;:=\; \log\frac{T^\sharp}{2\pi}-\frac{1}{T^\sharp}-A_L \;>\; 0$$

(hence $T^\sharp>T_1$). Then for every real even f with $\operatorname{supp} f\subseteq[-L,L]$,

$$Q(f) \;\ge\; R(f) \;:=\; 2F(i/2)^2+\frac{1}{\pi}\int_0^{T^\sharp}\big(\Psi_L(t)-\beta^*\big)|F(t)|^2\,dt+\beta^*\|f\|_2^2. \tag{6}$$

In the even Legendre basis of $L^2[-L,L]$ the form R is represented by $\beta^*I+2pp^{\mathsf T}+C$, where C has entries supported, up to super-exponentially small tails, on orders $n\lesssim eLT^\sharp/2$. Cutting after N even modes with first discarded order $2N\gtrsim eLT^\sharp/2$, if the leading block has $\lambda_{\min}\ge\lambda_0>0$, then

$$Q(f) \;\ge\; \big(\min(\lambda_0,\;\beta^*-\varepsilon_D)-\varepsilon_B\big)\,\|f\|_2^2$$

for all such f, where ε_D (tail deviation) and ε_B (leading–tail coupling) are explicit and, in the certified run below, both below 10^{-100}.

Proof. Split $\int_0^\infty=\int_0^{T^\sharp}+\int_{T^\sharp}^\infty$ in (2); for even real f, Parseval gives $\int_0^\infty|F|^2=\pi\|f\|^2$. On $[T^\sharp,\infty)$, Lemma 5 yields $\Psi_L(t)\ge\beta^*$, hence

$$\frac{1}{2\pi}\int_{|t|\ge T^\sharp}|F|^2\Psi_L \;\ge\; \beta^*\Big(\|f\|^2-\frac{1}{\pi}\int_0^{T^\sharp}|F|^2\Big),$$

which is (6). The operator C with symbol $(\Psi_L - \beta^*)\chi_{[0,T^\sharp]}$ has Legendre entries decaying super-exponentially once $n > eLT^\sharp/2$, by the spherical-Bessel bound $|j_n(x)| \le x^n/(2n+1)!!$. Writing the matrix of R in blocks $\left(\begin{smallmatrix} A & B \\ B^{\mathsf{T}} & D \end{smallmatrix}\right)$ after N even modes, the elementary estimate $\lambda_{\min}(R) \ge \min(\lambda_{\min}(A), \lambda_{\min}(D)) - \|B\|$ together with Gershgorin on D and a Schur bound on $\|B\|$ gives the last assertion. □

The frequency split is an exact direct sum: every discarded quantity is discarded with a definite sign, and the coupling to the tail is a single super-exponentially small norm. Order-space truncations of this problem leave $O(1)$ prime-shift coupling across the cut and fail.

The certificate at $L = 0.8$. The run uses $T^\sharp = 200$, $\beta^* = 0.5134\ldots$, $N = 200$ even Legendre modes, and 50-digit arithmetic. The leading block is assembled by Gauss–Legendre quadrature whose Bernstein-ellipse error is bounded by 10^{-42}; a verified Cholesky residual [16] at shift 8.9×10^{-18} succeeds with remainder below 10^{-50}. An independent run at $T^\sharp = 150$ gives a second, internally monotone certification. This is Theorem 1.

4 Parity

The restriction to even f can be removed. If $f = f_e + f_o$ is the even/odd splitting of a real test function supported in $[-L, L]$, then $Q(f) = Q(f_e) + Q(f_o)$, the pole term of $Q(f_o)$ changing sign to $-2\big(\int f_o \sinh(x/2)\,dx\big)^2$. For complex f one has $Q(f) = Q(\operatorname{Re} f) + Q(\operatorname{Im} f)$. Theorem 7 applies verbatim to each sector.

Theorem 8 (Simple even ground state, support 1.6)**.** *Let λ_1^{even}, λ_2^{even} be the first two min–max values of $Q/\|f\|_2^2$ over real even f with $\operatorname{supp} f \subseteq [-0.8, 0.8]$, and λ_1^{odd} the infimum over real odd f. Then*

$$8.9 \times 10^{-18} \;\le\; \lambda_1^{\text{even}} \;\le\; 2.523 \times 10^{-16}, \qquad \lambda_2^{\text{even}} \;\ge\; 2.085 \times 10^{-12},$$
$$8.206 \times 10^{-15} \;\le\; \lambda_1^{\text{odd}} \;\le\; 2.347 \times 10^{-14}.$$

The ground state is therefore simple and even, with clearance at least a factor 32 against the odd sector and 8×10^3 against the second even value.

Corollary 9. *For every complex $f \in L^2(\mathbb{R})$ with $\operatorname{supp} f \subseteq [-0.8, 0.8]$, $Q(f) \ge 8.9 \times 10^{-18}\,\|f\|_2^2$.*

The even-sector lower bound is Theorem 1; the odd-sector lower bound is the same reduction at $T^\sharp = 150$ with 200 odd Legendre modes; the even gap is a Courant–Fischer restriction test. The second missing step of [9]—convergence of approximate ground-state zeros to the zeta zeros as the scale grows—is untouched.

5 The Landau–Widom law

A function supported in $[-L, L]$ has Fourier transform of exponential type L; by Jensen, its real zeros have density at most L/π. The zeros of ζ have density $\nu(t) = \frac{1}{2\pi}\ln(t/2\pi)$, which reaches the budget L/π exactly at $T^*(L) = 2\pi e^{2L}$. Below T^* a test function can in principle interpolate every γ_j; above T^* the zeros outnumber the available oscillations. Killing $N = N(T^*)$ zeros with an exhausted budget is the Landau–Widom regime [13]: eigenvalues of a time–band limiting operator plunge at $\exp(-\pi^2 j/\ln K)$ past the Shannon number K. Modelling the suppression as an excursion of order N beyond Shannon suggests (4) rather than the naive "one nat per zero" guess $-\ln\lambda^* \asymp N(T^*)$.

L	n	dps	$\lambda^*(L) \le$	$-\ln\lambda^*$	$N(T^*)$	$\dfrac{-\ln\lambda^* \cdot \ln N}{N}$
0.5	24	90	1.05×10^{-6}	13.76	1	—
0.6	24	90	1.82×10^{-9}	20.13	1	—
0.7	24	90	4.99×10^{-13}	28.33	3	10.37
0.8	24	90	2.27×10^{-17}	38.32	4	13.28
0.9	31	95	5.91×10^{-23}	51.18	6	15.28
1.0	40	110	8.08×10^{-30}	66.99	8	17.42
1.1	53	110	2.78×10^{-38}	86.48	12	17.90
1.2	70	125	9.98×10^{-49}	110.53	16	19.16
1.4	119	155	1.91×10^{-77}	176.65	30	20.02
1.6	200	260	8.61×10^{-121}	276.46	55	20.15
1.8	333	420	7.86×10^{-186}	426.22	96	20.27
2.0	549	560	3.19×10^{-283}	650.47	165	20.13

Table 1: Unconditional certified upper bounds for $\lambda^*(L)$, from geometric-side interval evaluation of $Q(f_L)/\|f_L\|_2^2$. Here n is the number of odd sine modes, dps the geometric working precision, $N(T^*)$ the exact zero count below $T^* = 2\pi e^{2L}$. The last column plateaus at $20.1 \approx 2\pi^2 = 19.74$.

Each entry of Table 1 is the interval-arithmetic upper endpoint of a geometric-side Rayleigh quotient for an explicit trial function f_L in a sine basis of $L^2(-L, L)$. The assembly involves only a finite prime sum, a compact archimedean integral, and an explicit pole. A zeros-side pipeline proposes the coefficients of f_L; the origin of a trial vector needs no certification for a variational upper bound to be valid.

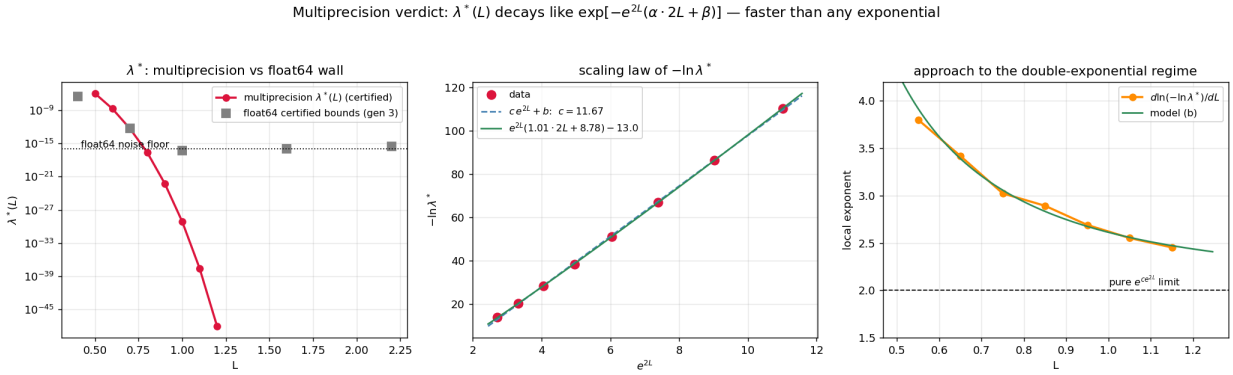


Figure 1: Certified upper bounds for $L \le 1.2$. Left: the bounds fall through the float64 wall. Centre: $-\ln\lambda^*$ against e^{2L}. Right: the local exponent $d\ln(-\ln\lambda^*)/dL$ decreasing towards its asymptote.

The per-zero cost $-\ln\lambda^*/N(T^*)$ decreases steadily, excluding a law proportional to $N(T^*)$. Multiplied by $\ln N(T^*)$ it plateaus: 20.02, 20.15, 20.27, 20.13 at the last four points, about 2% above $2\pi^2$. A single-constant fit to those four points gives $C = 20.13 \pm 0.10$, with in-sample residuals below 0.7%; the parameter-free value $C = 2\pi^2$ reproduces them to within 1.4%–2.6%. Out of sample, models calibrated on $L \le 1.2$ are required to predict $L = 1.4, 1.6, 1.8, 2.0$. The Landau–Widom form underpredicts by a nearly constant factor $(-4.4\%, -4.9\%, -5.5\%, -4.8\%)$, with no trend: the offset is finite-size drift of a constant that has not yet saturated at $L \le 1.2$. The competing three-parameter model $-\ln\lambda^* = e^{2L}(\alpha \cdot 2L + \beta) + \delta_0$ errs instead by $+0.8\%, +2.0\%, +3.7\%, +5.8\%$, monotonically growing, which is the signature of a wrong functional form.

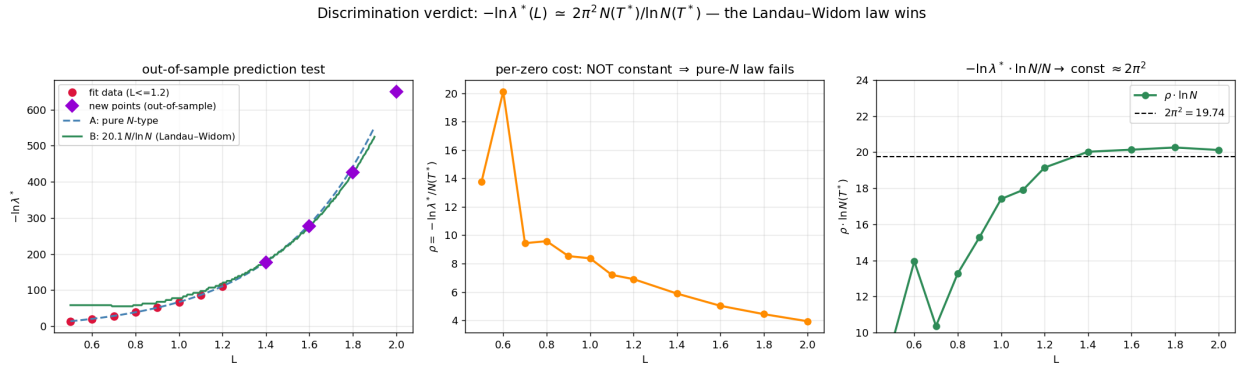


Figure 2: Model discrimination. Left: out-of-sample prediction of $L = 1.4, 1.6, 1.8$ by the two laws calibrated on $L \le 1.2$. Centre: the per-zero cost $-\ln\lambda^*/N(T^*)$ is not constant. Right: multiplied by $\ln N(T^*)$ it plateaus at $20.1 \approx 2\pi^2$.

What is certified is that each displayed value is an *upper* bound for $\lambda^*(L)$. Identifying (4) with λ^* itself rests on completeness of the sine basis (Galerkin values decrease to the infimum; we observe stabilization at the 0.03% level) and on the two-sided enclosure at $L = 0.8$. We formalize the identification as:

Conjecture 10. *As $L \to \infty$,*

$$-\ln\lambda^*(L) \;=\; 2\pi^2 \,\frac{N(T^*)}{\ln N(T^*)}\,\big(1 + o(1)\big), \qquad T^* = 2\pi e^{2L}.$$

The shape $N/\ln N$ is what the data discriminate; the constant $2\pi^2$ is read off the plateau. On the certified range the two normalizations $2\pi^2\, N/\ln N$ and $4\pi\, e^{2L}$ are entangled and cross near $L \approx 2.9$. We state Conjecture 10 as the reading of the present data.

6 A conditional upper bound

Theorem 2 captures the qualitative content of (4)—decay faster than exponential, at least double-exponential with exponent e^L—but neither the scale $T^* = 2\pi e^{2L}$ nor the constant $2\pi^2$. The construction operates at half saturation, and that is what limits the exponent.

Assume RH, and set $T = 2\pi e^L$. Let

$$\hat{f}_0(r) \;=\; B(r)\,h(r), \qquad B(r) = \prod_{0<\gamma_j\le T}\Big(1 - \frac{r^2}{\gamma_j^2}\Big), \qquad h(r) = \Big(\frac{\sin(Lr/m)}{Lr/m}\Big)^m,$$

with $m = \lceil LT/e \rceil$. Then B annihilates every zero up to T, h has exponential type exactly L, and $m > 2N(T) + 2$ for large L, so $\hat{f}_0 \in L^1 \cap L^2$ and $f_0 = \mathcal{F}^{-1}\hat{f}_0$ is admissible of support $[-L, L]$. Under RH, $Q(f_0) = \sum_{\gamma>T} 2\,\hat{f}_0(\gamma)^2$. A potential-theoretic bound on $\log|B|$ along the real axis, balanced against the power-of-sinc decay of h, shows that the Rayleigh quotient is $\le \exp(-Le^L)$ for large L. The obstruction to reaching the full exponent e^{2L} is an extremal problem for band-limited windows, the same problem whose solution is governed by Landau–Widom asymptotics.

7 The barrier, and a closing remark

Theorem 3 is immediate from Theorem 7 and Lemma 6: $\beta^* > 0$ forces $T^\sharp > T_1 = 2\pi e^{A_L}$, the PNT gives $A_L \sim 4e^L$, and no pointwise bound on the comb can do better. At $L = 1.6$ one already has $T_1 \approx 10^7$; past support ≈ 3.2 a pointwise-envelope certificate is computationally void.

Two heights coexist. Defeating positivity—finding small Rayleigh quotients—is governed by the resolution height $T^* = 2\pi e^{2L}$, singly exponential, because the minimizer only needs to interpolate the zeros it can resolve. Certifying positivity through the geometric side is governed by $T_1 = 2\pi \exp\big((4 + o(1))e^L\big)$, doubly exponential, because the certificate must exclude worst-case phase alignment of the prime comb. The gap between e^{2L} and e^{4e^L} is the gap between knowing where the zeros are and proving that the primes cannot conspire.

Certificate, law and barrier are what we take this program to deliver: the frontier of unconditional positivity moves from $\log 2$ to 1.6, the profile beyond it is measured out to $L = 2$, and the wall behind both is mapped rather than left implicit.

References

[1] A. Weil, *Sur les "formules explicites" de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund, Tome Supplémentaire (1952), 252–265.

[2] K. Barner, *On A. Weil's explicit formula*, J. reine angew. Math. **323** (1981), 139–152.

[3] H. Yoshida, *On Hermitian forms attached to zeta functions*, in: Zeta Functions in Geometry, Adv. Stud. Pure Math. **21**, 1992, 281–325.

[4] E. Bombieri, *Remarks on Weil's quadratic functional in the theory of prime numbers, I*, Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei (9) Mat. Appl. **11** (2000), 183–233.

[5] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), 29–106.

[6] A. Connes and C. Consani, *Weil positivity and trace formula, the archimedean place*, Selecta Math. (N.S.) **27** (2021), no. 4, Paper No. 77.

[7] A. Connes and H. Moscovici, *The UV prolate spectrum matches the zeros of zeta*, Proc. Natl. Acad. Sci. USA **119** (2022), no. 22, e2123174119.

[8] A. Connes and W. D. van Suijlekom, *Quadratic forms, real zeros and echoes of the spectral action*, preprint, arXiv:2511.23257 (2025).

[9] A. Connes, C. Consani and H. Moscovici, *Zeta spectral triples*, preprint, arXiv:2511.22755 (2025).

[10] M. Suzuki, *Weil's quadratic form via the screw function*, preprint, arXiv:2606.09096 (2026).

[11] A. Groskin, *A finite Guinand–Weil dictionary and archimedean tail order for the truncated Weil quadratic form*, preprint, arXiv:2607.02828 (2026).

[12] D. Slepian and H. O. Pollak, *Prolate spheroidal wave functions, Fourier analysis and uncertainty, I*, Bell System Tech. J. **40** (1961), 43–63.

[13] H. J. Landau and H. Widom, *Eigenvalue distribution of time and frequency limiting*, J. Math. Anal. Appl. **77** (1980), 469–481.

[14] B. Rodgers and T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum Math. Pi **8** (2020), e6.

[15] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloq. Publ. **53**, 2004.

[16] S. M. Rump, *Verification methods: rigorous results using floating-point arithmetic*, Acta Numerica **19** (2010), 287–449.

[17] K. Seip, *Interpolation and Sampling in Spaces of Analytic Functions*, University Lecture Series **33**, American Mathematical Society, 2004.